

Admixture-geometry exposure diagnostic

Does near-fixation arise from admixture geometry rather than sorting?

Question

Units are retained at pooled parental minor-allele frequency (MAF) ≥ 0.15 , so the two parental species need not differ by a fixed allele. In principle a unit could clear the near-fixation floor ϕ in a hybrid population through *admixture geometry alone*—no sorting—if the population’s ancestry is skewed and the parental frequencies are lopsided in the same direction. Because the reported sorting *direction* depends on the diagnostic index (DI), a DI-structured geometric artefact would be a competing explanation for that result. This is a diagnostic: no null model, drift correction, or significance test.

Method

For each unit i and hybrid population k the expected *F. aquilonia*-allele frequency under pure admixture is

$$\tilde{q}_{ik} = h_k p_i^{\text{aqu}} + (1 - h_k) p_i^{\text{pol}},$$

where $p_i^{\text{aqu}}, p_i^{\text{pol}}$ are the oriented parental frequencies and h_k is the genome-wide hybrid index. At $h = 0.5$ this reduces to the pooled parental frequency $\bar{p} = (p^{\text{aqu}} + p^{\text{pol}})/2$, which the MAF filter confines to $[0.15, 0.85]$; geometry therefore cannot clear $\phi = 0.85$ at a balanced hybrid index and is strictly excluded at 0.90 and 0.95. A cell is *exposed* when $\tilde{q}_{ik} \geq \phi$ (or $\leq 1 - \phi$). Orientation follows the sorting pipeline (higher oriented frequency = *F. aquilonia* allele). No pre-computed hybrid index exists in the data, so h_k is derived from the *same* oriented frequencies by least-squares regression of hybrid on parental frequencies—making an orientation sign error impossible by construction. The retained set is 658,057 differentiated units across 20 hybrid populations; $\phi \in \{0.80, 0.85, 0.90, 0.95\}$.

Across populations h ranges 0.34 (Åland) to 0.62 (Heinamäki), most within 0.1 of 0.5: these are near-balanced hybrids, leaving little room for geometric exposure.

Result

The key quantity is the fraction of *observed* near-fixed calls (the actual sorting signal) that a cell’s geometry would also predict.

At $\phi \geq 0.90$ exposure is **exactly zero in every decile**. At the primary $\phi = 0.85$ it is $\sim 5\%$ overall and—decisively—*humped in the middle deciles* (5–7, where sorting direction is balanced near 0.5) and *lowest in exactly the deciles that drive the reported reversal*: the two lowest DI deciles (2–3%) and the four highest, falling $7.5\% \rightarrow 0.4\%$. Directionally, the high-DI $\rightarrow$ *F. aquilonia* calls that anchor the reversal are essentially unexposed (0.7% at decile 9, 0.4% at decile 10; the exposed *F. aquilonia* calls peak at 11.7% in the balanced decile 6).

Validation. All checks pass. Forcing $h = 0.5$ gives exactly zero exposed cells for $\phi > 0.85$ (and only the $\bar{p}$ boundary at 0.85), confirming orientation and machinery. Exposed cells concentrate in the most ancestry-skewed populations (per-population exposed cells vs. $|h - 0.5|$, Spearman

Fraction of observed near-fixed calls explainable by admixture geometry, by DI decile $\times \phi$.

DI decile	$\phi = 0.80$	$\phi = 0.85$	$\phi = 0.90$	$\phi = 0.95$
1 (low)	0.147	0.021	0	0
2	0.228	0.033	0	0
3	0.331	0.057	0	0
4	0.338	0.061	0	0
5	0.358	0.081	0	0
6	0.335	0.084	0	0
7	0.300	0.075	0	0
8	0.195	0.044	0	0
9	0.077	0.019	0	0
10 (high)	0.016	0.004	0	0
overall	0.242	0.051	0	0

$\rho = 0.87$; Aland, the most skewed at $h = 0.34$, contributes the most). $\tilde{q} \in [0, 1]$ throughout, and the retained set satisfies parental MAF ≥ 0.15 .

Discreteness sub-check. Near-fixation is $\hat{q} \geq \phi$ on $2n_k$ gametes, and populations run $n_k = 3$ –20 diploids, so the effective bar varies: at $n_k = 3$ it is $6/6 = 1.0$, at $n_k = 4$ it is $7/8 = 0.875$, falling to 0.85 for $n_k \geq 10$. Despite this, observed near-fixed rates are similar across sizes (0.34–0.52), and small populations ($n_k \leq 5$) contribute a *flat* 31–35% of calls across all DI deciles—no DI-structured distortion.

Conclusion

Admixture geometry does not reproduce the DI structure of sorting direction. At the primary floor only $\sim 5\%$ of near-fixed calls are geometry-compatible, concentrated where direction is balanced and negligible in the reversal-driving deciles; at $\phi \geq 0.90$ geometric exposure is zero, so the DI-dependent reversal—which persists at those stricter floors—cannot be a geometric artefact. The competing explanation is retired; no re-analysis is warranted.

Diagnostic script: `moduleA_sorting/R/moduleA_admixture_exposure.R`
 Cached results: `moduleA_sorting/data/moduleA_admixture_exposure.rds`
 Retained set 658,057 differentiated units $\times$ 20 hybrid populations.